EE3032 Introduction to VLSI Design – Final

Fall 2024

1. (45 points) Answer the following questions.

(a) List the components of dynamic power consumption of a dynamic logic gate. (6 points)

(b) The power integrity (IR drop) of an IC is related to the design of power distribution network. List two approaches to improve the power integrity. (6 points)

(c) Explain the latchup phenomenon. List a design approach which can alleviate the occurrence probability of latchup. (6 points)

(d) Negative bias temperature instability (NBTI) is one of primary mechanisms for oxide wearout. Explain the principle of NBTI. (4 points)

(e) List two design approaches for overcome the issue of hold time violation in a shift register. (6 points)

(f) List two requirements for drawing the layout of a cell in the standard cell library. (4 points)

(g) List two methods to reduce the power bounce and ground bounce occurring in the IO circuits. (4 points)

(h) List two design methods for avoiding glitch power of a CMOS circuit. (4 points)

(i) Design $f = \overline{x_1 x_2 + x_3}$ using differential pass transistor logic. Draw the transistor-level schematic diagram. (5 points)

2. (5 points) If we want to design a 3-bit adder, we can use the carry generation function $c_{i+1} = p_i c_i + g_i$ for $i = 0, 1, 2$ to generate the carries c_1, c_2, c_3 of the adder. Design a carry generation circuit with inputs $c_0, p_0, p_1, p_2, g_0, g_1, g_2$ for a 3-bit adder using multioutput domino logic. Draw the transistor-level schematic diagram.

3. (15 points) Fig. 1 shows a simple circuit with three gates. Complete the following questions.

(a) Can the circuit be operated correctly? Explain your answer. If your answer is "no", modify the circuit such that it can be operated correctly. (5 points)

(b) Assume that each transistor of the (modified) circuit has the following drain and gate capacitance: $C_d = 0.1\,\text{fF}$, $C_g = 1\,\text{fF}$; and the parasitic capacitance of the node x and y are 1.3 fF and 1.1 fF, respectively. Let the switching voltage of all the gates in the (modified) circuit be $V_{DD}/2$. Can the circuit be operated correctly? Explain your answer. (5 points)

(c) The clock period of the (modified) circuit cannot be prolonged casually. Why? Modify the circuit such that its clock period can be prolonged casually. Draw the transistor schematic diagram. (5 points)

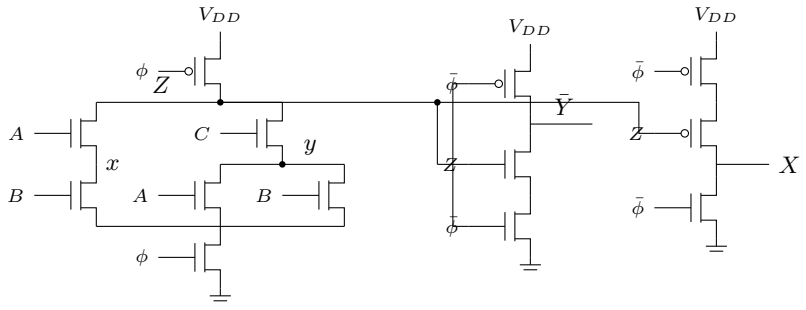
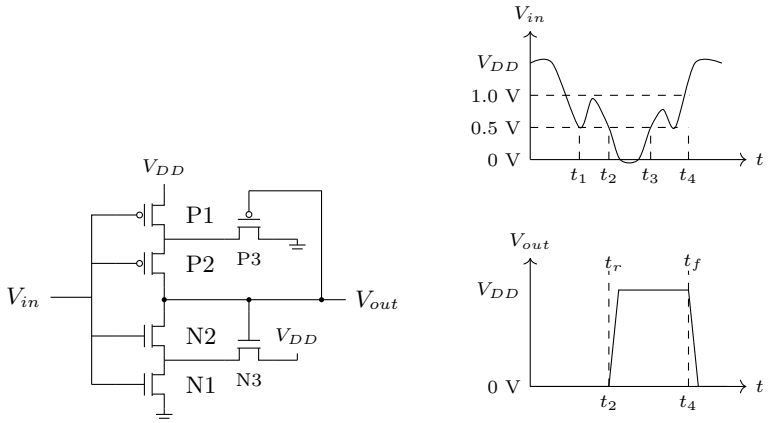


Figure 1: A simple circuit.

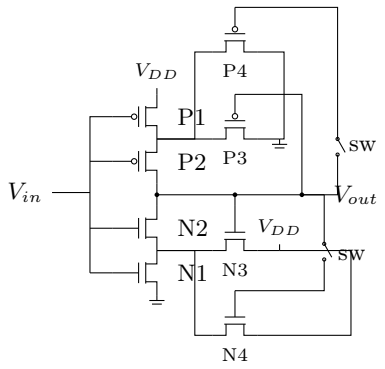
4. (11 points) Fig. 2(a) shows a Schmitt trigger circuit. If we apply an input signal as shown in the left figure to the Schmitt trigger circuit, the output signal will be as the right figure shows. Assume that $\mu_n = 2\mu_p$, the drain parasitic capacitance of all the transistors can be ignored, and the simple RC delay model is used to estimate the delay. Complete the following questions.

(a) Let $V_{tn} = |V_{tp}| = 0.2\,\text{V}$ and $V_{DD} = 1.5\,\text{V}$. If the size of N1 is 160 nm/40 nm, determine the sizes of the transistors N2, N3, P1, P2 and P3 such that $t_r = t_f = 3RC$, where the turn-on resistance and gate capacitance of N1 is R and C. All the transistors should have the same channel length. (5 points)

(b) Fig. 2(b) shows Schmitt trigger circuit (STC) with controllable hysteresis voltage (V_H) by controlling the switches (sw). If the switches are turned on, is the noise tolerance capability of the STC increased or decreased? Is the short-circuit power of STC increased or decreased? Explain your answers. (6 points)



(a)



(b)

Figure 2: (a) A CMOS Schmitt trigger circuit (b) A STC with controllable hysteresis voltage.

5. (10 points) Draw the transistor-level schematic diagram of the symbolic layout shown in Fig. 3(a).

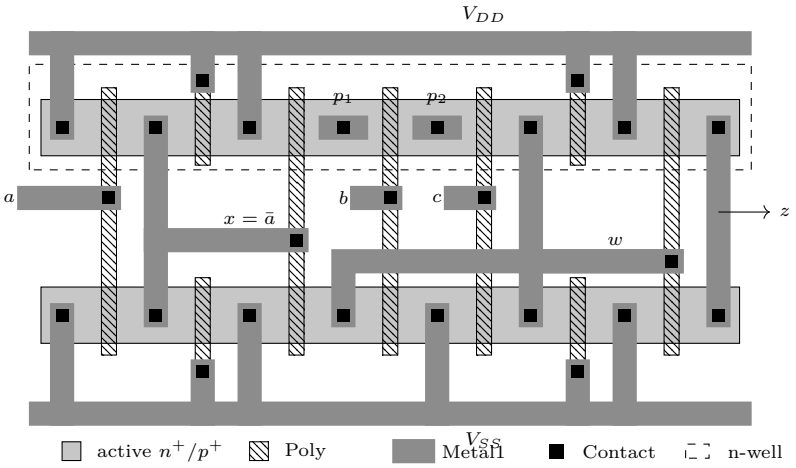


Figure 3(a): Symbolic layout.

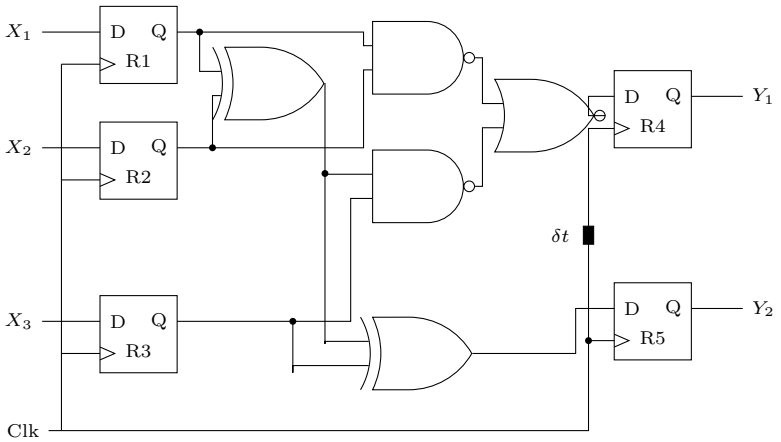


Figure 3(b): A sequential circuit.

6. (24 points) Fig. 3(b) shows a sequential circuit. Assume that the clock-to-Q propagation delay (t_{pcq}), the clock-to-Q contamination delay (t_{ccq}), the setup time (t_{setup}), and the hold time (t_{hold}) of the registers are 0.2 ns, 0.1 ns, 0.2 ns, and 0.2 ns, respectively. Also, $\delta t = 0.6$ ns, $t_{NAND} = 0.5$ ns, $t_{NOR} = 0.5$ ns, and $t_{XOR} = 1$ ns. Complete the following questions.

(a) Determine the maximum operation frequency of the circuit. (5 points)

(b) Assume that the delay of an inverter is 0.3 ns. Redesign the clock tree by doing buffer insertion such that the maximum operation frequency can be boosted to 400 MHz. Draw the clock tree including the registers. (5 points)

(c) Design the combinational circuit using NP-domino logic style. Draw the transistor-level schematic diagram. (10 points)

(d) In the two registers, R4 and R5, which register has higher soft error rate? Explain your answer. (4 points)

Jin-Fu Li, EE, NCU. Typeset transcription; all figures are editable vectors. The middle clock overbar touches a wire in the scan; its phase readings are discussed in the solution. The original printed point allocations sum to 110; they are retained without rescaling. Original annotations and photo perspective are excluded.